Development of AI Response Validation System with Hallucination Detection Assistance

Evaluation Report Export — Milestone 4

Project Title	Development of AI Response Validation System with Hallucination Detection Assistance
Group	Group 1
Milestone	Milestone 4
Report Type	Batch Evaluation Report
Format	PDF
Total Evaluations	10

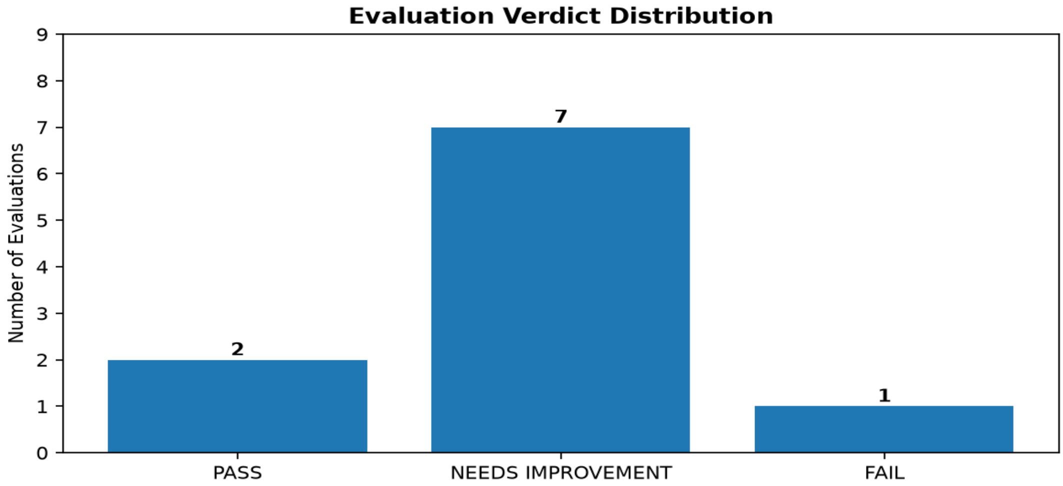
1. Batch Evaluation Summary

Metric	Result
Total Evaluations	10
PASS	2
NEEDS IMPROVEMENT	7
FAIL	1
Average Overall Score	7.43/10
Hallucination Frequency	10.0%

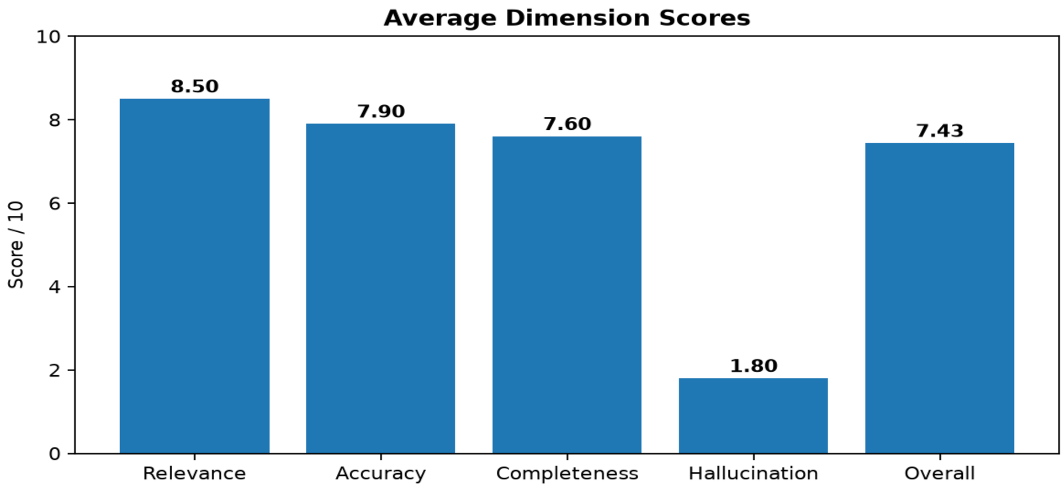
2. Dimension-wise Score Analysis

Dimension	Average Score
Relevance	8.50/10
Accuracy	7.90/10
Completeness	7.60/10
Hallucination	1.80/10
Overall Score	7.43/10

3. Evaluation Charts

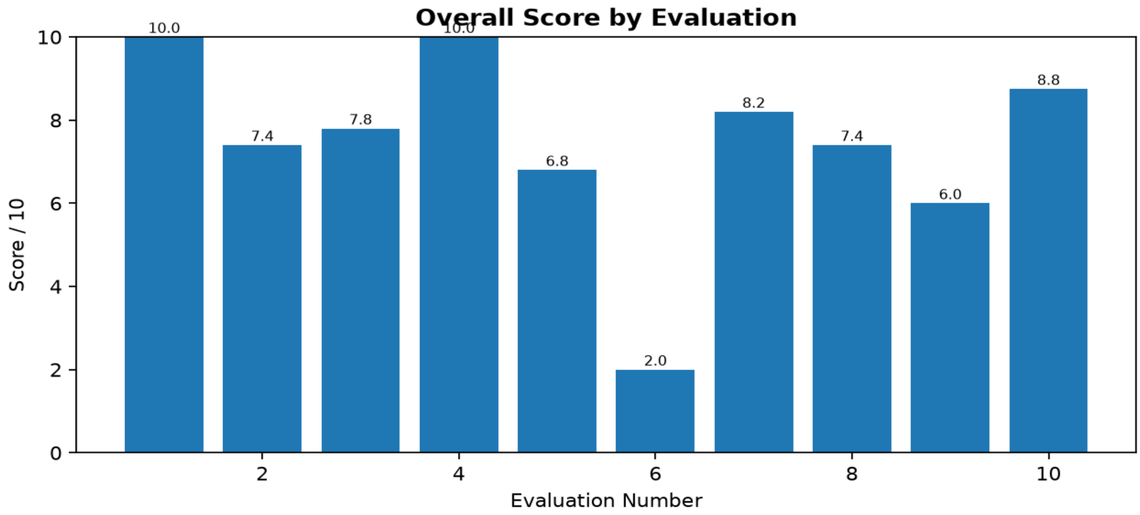


The chart shows the distribution of PASS, NEEDS IMPROVEMENT, and FAIL verdicts in the evaluation batch.



The chart compares the average scores across the evaluation dimensions.

4. Overall Score Trend



The chart shows the overall score obtained for each evaluation in the batch.

5. Individual Evaluation Results

Evaluation 1

Question: What is Artificial Intelligence?

AI Response: Artificial Intelligence is the field of creating systems that can perform tasks requiring human intelligence, such as learning, reasoning, and problem solving.

Relevance	Accuracy	Completeness	Hallucination	Overall
10.0/10	10.0/10	10.0/10	0.0/10	10.0/10

Overall Verdict: PASS

Evaluation 2

Question: What is Machine Learning?

AI Response: Machine Learning is a subset of AI that allows computers to learn patterns from data.

Relevance	Accuracy	Completeness	Hallucination	Overall
9.0/10	8.0/10	8.0/10	2.0/10	7.4/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 3

Question: What is Deep Learning?

AI Response: Deep Learning uses neural networks with multiple layers to learn complex patterns from data.

Relevance	Accuracy	Completeness	Hallucination	Overall
9.0/10	8.0/10	8.0/10	2.0/10	7.8/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 4

Question: What is Python?

AI Response: Python is a high-level, interpreted programming language known for its simple syntax, readability, and wide use in software development, data science, and AI.

Relevance	Accuracy	Completeness	Hallucination	Overall
10.0/10	10.0/10	10.0/10	0.0/10	10.0/10

Overall Verdict: PASS

Evaluation 5

Question: What is a neural network?

AI Response: A neural network is a machine learning model made of interconnected nodes arranged in layers.

Relevance	Accuracy	Completeness	Hallucination	Overall
9.0/10	8.0/10	8.0/10	2.0/10	6.8/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 6

Question: What is supervised learning?

AI Response: it is type of ai.

Relevance	Accuracy	Completeness	Hallucination	Overall
2.0/10	2.0/10	2.0/10	8.0/10	2.0/10

Overall Verdict: FAIL

Evaluation 7

Question: What is unsupervised learning?

AI Response: Unsupervised learning finds patterns or groups in data without labeled outputs.

Relevance	Accuracy	Completeness	Hallucination	Overall
9.0/10	8.0/10	8.0/10	0.0/10	8.2/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 8

Question: What is overfitting in machine learning?
AI Response: Overfitting happens when a model learns the training data too closely and performs poorly on new data.

Relevance	Accuracy	Completeness	Hallucination	Overall
9.0/10	8.0/10	8.0/10	2.0/10	7.4/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 9

Question: What is Natural Language Processing?
AI Response: NLP helps computers process and understand human language.

Relevance	Accuracy	Completeness	Hallucination	Overall
8.0/10	8.0/10	6.0/10	2.0/10	6.0/10

Overall Verdict: NEEDS IMPROVEMENT

Evaluation 10

Question: What is an AI hallucination?
AI Response: An AI hallucination occurs when an AI generates false or unsupported information as if it were factual.

Relevance	Accuracy	Completeness	Hallucination	Overall
10.0/10	9.0/10	8.0/10	0.0/10	8.8/10

Overall Verdict: NEEDS IMPROVEMENT

6. Hallucinated Responses

Evaluation 6

Question: What is supervised learning?
Hallucination Score: 8.0/10
Verdict: FAIL

7. Flagged Responses

Evaluation	Overall	Verdict	Hallucination	Reason
2	7.4/10	NEEDS IMPROVEMENT	2.0/10	Needs improvement
3	7.8/10	NEEDS IMPROVEMENT	2.0/10	Needs improvement
5	6.8/10	NEEDS IMPROVEMENT	2.0/10	Needs improvement
6	2.0/10	FAIL	8.0/10	FAIL, Hallucination flag
7	8.2/10	NEEDS IMPROVEMENT	0.0/10	Needs improvement
8	7.4/10	NEEDS IMPROVEMENT	2.0/10	Needs improvement
9	6.0/10	NEEDS IMPROVEMENT	2.0/10	Needs improvement
10	8.8/10	NEEDS IMPROVEMENT	0.0/10	Needs improvement

8. Overall Verdicts

Verdict	Count	Percentage
PASS	2	20.0%
NEEDS IMPROVEMENT	7	70.0%
FAIL	1	10.0%

The batch contains 2 PASS, 7 NEEDS IMPROVEMENT, and 1 FAIL evaluations.

9. Improvement Recommendations

1. Review the 1 failed response(s) and identify the recurring causes of poor evaluation scores.

2. Review the 7 responses marked NEEDS IMPROVEMENT and improve their weak dimensions.

Conclusion: This report provides a structured summary of the batch evaluation, including project details, evaluation results, dimension-wise scores, hallucination analysis, flagged responses, verdict distribution, charts, and improvement recommendations.